

Attention Is All You Need (Summary)

The Transformer is a model architecture that relies entirely on an attention mechanism to draw global dependencies between input and output, dispensing with recurrence and convolutions entirely. This allows for significantly more parallelization during training.

Self-Attention

Self-attention relates different positions of a single sequence in order to compute a representation of that sequence. The scaled dot-product attention computes attention weights as a softmax over the dot products of queries and keys, scaled by the square root of the key dimension.

Multi-Head Attention

Instead of performing a single attention function, the Transformer projects the queries, keys, and values h times with different learned projections. This lets the model jointly attend to information from different representation subspaces at different positions.